QUILT-1M: One Million Image-Text Pairs for Histopathology

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- Equal Contribution

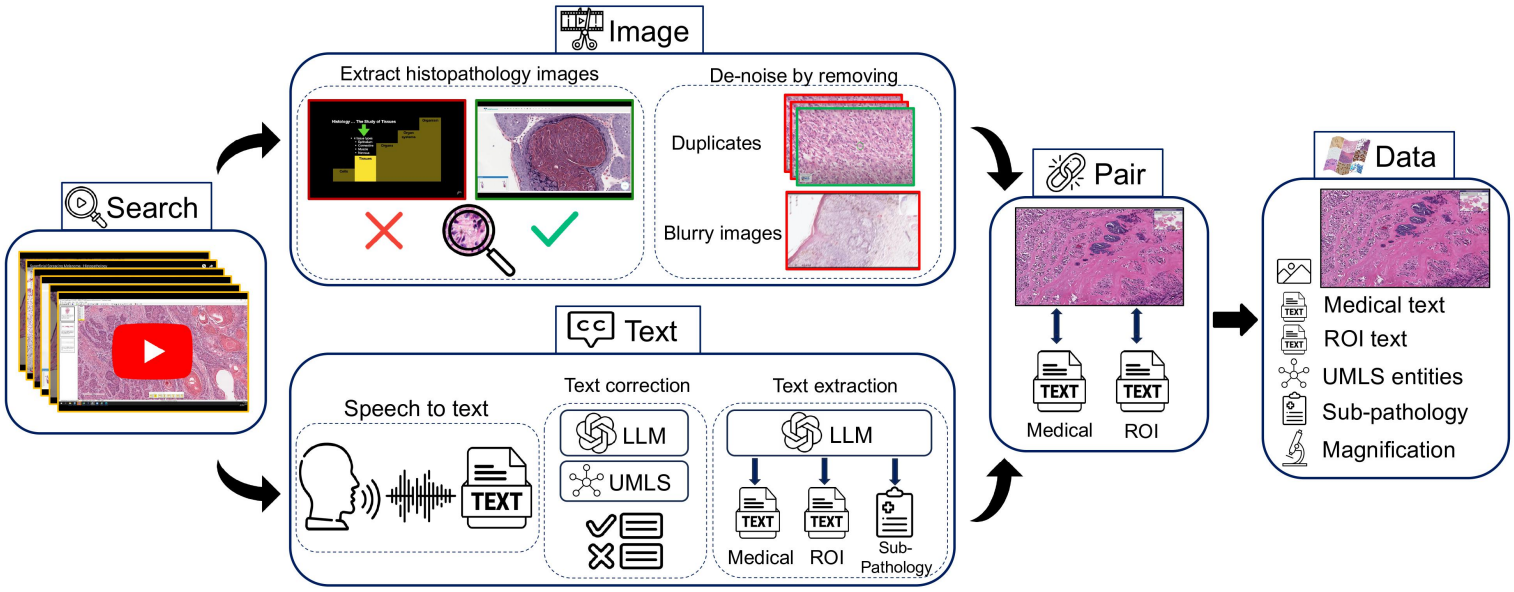
Paper (https://arxiv.org/abs/2306.11207) Code (https://github.com/wisdomikezogwo/quilt1m)

Data (https://github.com/wisdomikezogwo/quilt1m) Model/ Demo (https://huggingface.co/wisdomik/QuiltNet-B-32)

Abstract

Recent accelerations in multi-modal applications have been made possible with the plethora of image and text data available online. However, the scarcity of similar data in the medical field, specifically in histopathology, has halted similar progress. To enable similar representation learning for histopathology, we turn to YouTube, an untapped resource of videos, offering **1,087** hours of valuable educational histopathology videos from expert clinicians. From YouTube, we curate QUILT: a large-scale vision-language dataset consisting of **768,826** image and text pairs. Quilt was automatically curated using a mixture of models, including large language models), handcrafted algorithms, human knowledge databases, and automatic speech recognition. In comparison, the most comprehensive datasets curated for histopathology amass only around **200K** samples. We combine Quilt with datasets, from other sources, including Twitter, research papers, and the internet in general, to create an even larger dataset: QUILT-1M, with **1**M paired image-text samples, marking it as the largest vision-language histopathology dataset to date. We demonstrate the value of QUILT-1M by fine-tuning a pre-trained CLIP model. Our model outperforms state-of-the-art models on both zero-shot and linear probing tasks for classifying new pathology images across **13** diverse patch-level datasets of **8** different sub-pathologies and cross-modal retrieval tasks.

Model and Objectives



To address the need for a large-scale vision-language dataset in histopathology, we introduce QUILT: containing 419,780 images aligned with 768,826 text pairs. We draw on the insight that publicly available educational YouTube histopathology content represents an untapped potential. We curate QUILT using 1,087 hours of valuable educational histopathology videos from expert pathologists on YouTube. To extract aligned image and text pairs from the videos, we utilize a mixture of models: large language models, handcrafted algorithms, human knowledge databases, and automatic speech recognition. QUILT does not overlap with any current open-access histopathology data sources. This allows us to merge our dataset with other open-source datasets available. Therefore, to create an even larger and more diverse dataset, we combine QUILT with data from other sources, such as Twitter, research papers, and the Internet, resulting in QUILT-1M. The larger QUILT-1M contains one million image-text pairs, making it the largest public vision-language histopathology dataset to date.

Dataset

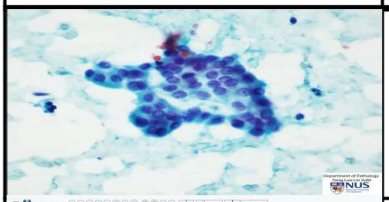
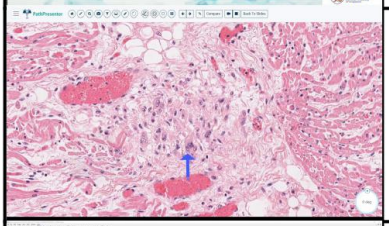
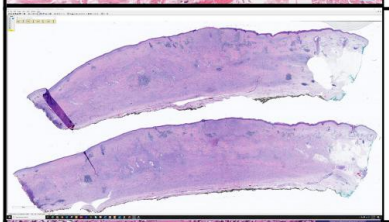
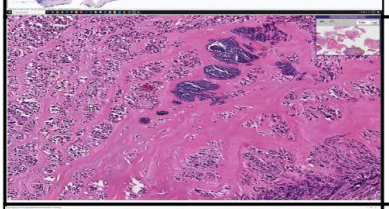
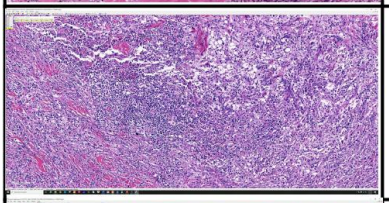
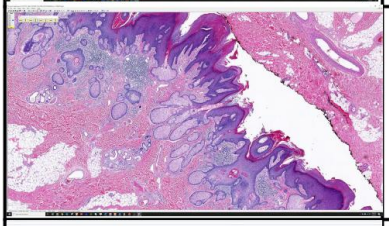
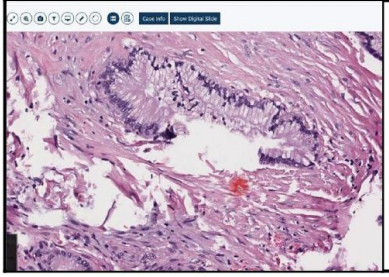
We collected QUILT, from 4504 narrative videos spanning over 1087 hours with over 438K unique images with 768K associated text pairs. The mean length of the text captions is 22.76 words, and 8.68 words for ROI text, with an average of 1.74 medical sentences per image (max=5.33, min=1.0). Our dataset spans a total of 1.469M UMLS entities from those mentioned in the text (with 28.5K unique). The images span varying microscopic magnification scales (0-10x, 10-20x, 20-40x), obtaining (280K, 75K, 107K) images from each scale respectively.

One of our narrative videos can be seen below.

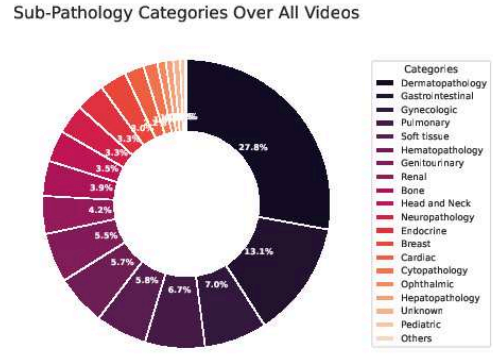
Filip Sokol

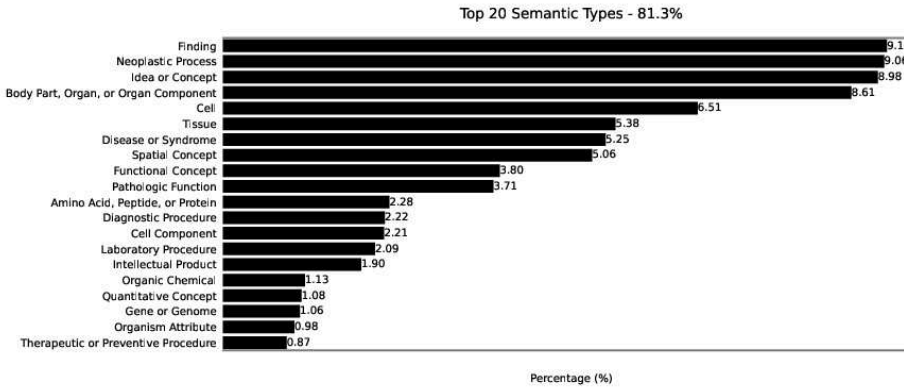
Watch on

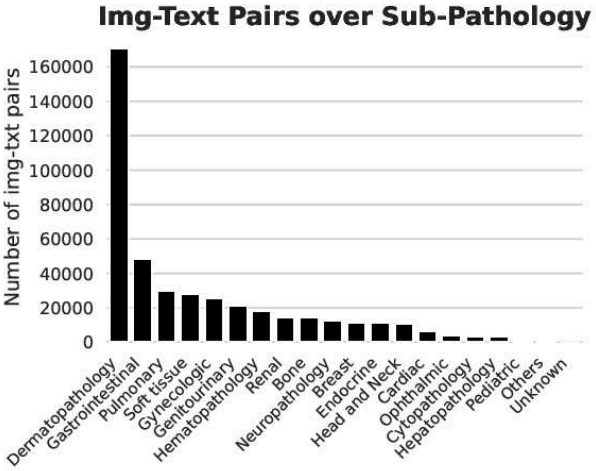
Following are the collection of sample images from our dataset, accompanied by corresponding medical text, ROI text, and the top three sub-pathology classifications:

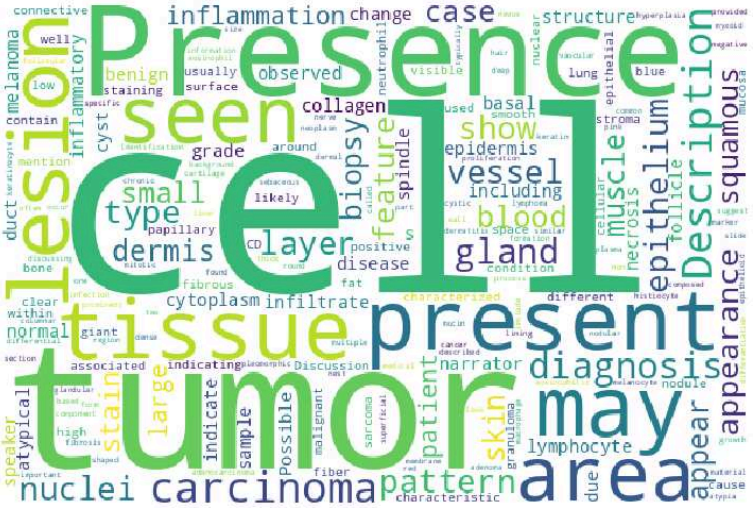
Image	Medical TEXT	ROI Text	Sub-pathology Classification
	['There are clusters of cells with micro-follicular formations.','Nuclear pseudo-inclusions, oval nuclei, nuclear grooves, and small nucleoli are present in some cells.']	['clusters of cells', 'micro-follicular formations', 'nuclear pseudo-inclusions', 'oval nuclei', 'nuclear grooves', 'small nucleoli']	['Endocrine', 'Cytopathology', 'Head and Neck']
	['Cluster of macrophages and T cells is characteristic of acute rheumatic fever.', 'Aschoff body is a characteristic feature of acute rheumatic fever.', 'Macrophages with elongated chromatin are called Anitchkow cells and are commonly seen in Aschoff bodies.', 'Pancarditis with Aschoff bodies is present.']	['Cluster of macrophages and T cells', 'Aschoff body', 'Macrophages with elongated chromatin', 'Anitchkow cells', 'Pancarditis']	['Cardiac', 'Hematopathology', 'Endocrine']
	['An 80-year-old man has a scar-like plaque on the scalp that has been called malignant on a biopsy.', 'The tissue affected by the plaque extends from the epidermis to the galea aponeurotica, near the periosteum of the skull.', 'The skin, dermis, and subcutis are all affected by the process.']	['scar-like plaque on the scalp', 'malignant on a biopsy', 'skin, dermis, and subcutis affected by the process']	['Dermatopathology', 'Soft tissue', 'Hematopathology']
	['Inflammatory cells surrounding cartilage can indicate acute chondritis, with neutrophils being the principal cell type.', 'Chronic chondritis may be diagnosed if lymphocytes are the predominant inflammatory cell type.']	['cartilage', 'inflammatory cells']	['Hematopathology', 'Bone', 'Dermatopathology']
	['Large histiocytes with abundant cytoplasm identified as Rosai-Dorfman histiocytes.', 'S100 stain showed perivascular cuffing.', 'Initial diagnosis of inflammatory pseudotumor of the orbit.', 'Rosai-Dorfman disease may burn out and leave behind fibrotic pockets.']	['Large histiocytes', 'perivascular cuffing', 'fibrotic pockets'	['Dermatopathology', 'Soft tissue', 'Hematopathology']
	['Epidermal acanthosis and papillomatosis resembling a wart or seborrheic keratosis.', 'Presence of large sebaceous glands that drain directly through their duct out to the skin surface, which is abnormal.', 'Presence of a demodex mite.']	['Epidermal acanthosis and papillomatosis', 'large sebaceous glands', 'demodex mite']	['Dermatopathology', 'Soft tissue', 'Hematopathology']
	['Histological description of glandular tissue with little atypia but located in a place where it does not belong can be a helpful criteria to discern the presence of malignancy.', 'Glands located on the periphery and infiltrating into adventitia and peripancreatic tissue may be malignant.']	['glandular tissue', 'pancreas',]	['Gastrointestinal', 'Pancreatic', 'Hematopathology']

Following figures illustrate the distribution of data across 18 sub-pathology types, offering a comprehensive analysis of the dataset's text distribution.



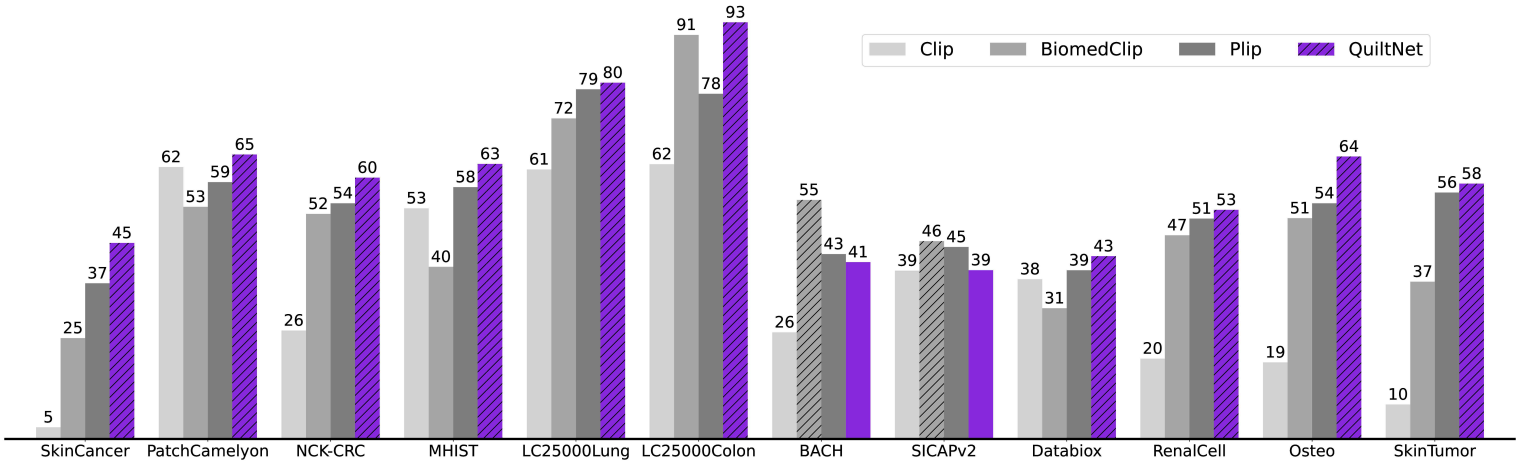






Results

We use the Contrastive Language-Image Pre-training (CLIP) objective (https://proceedings.mlr.press/v139/radford21a.html) to pretrain QuiltNet using Quilt-1M. QuiltNet, outperforms out-of-domain CLIP baseline and state-of-the-art histopathology models across 12 zero-shot tasks, covering 8 different sub-pathologies (accuracy percentage provided).



Results using linear probing: We assess the few-shot and full-shot performance of our model by conducting linear probing with 1%, 10%, and 100% of the training data, sampled with three different seeds. Remarkably, our model, utilizing the ViT-B/32 (https://openreview.net/forum?id=YicbFdNTTy) architecture with GPT/77 (https://www.semanticscholar.org/paper/Language-Models-are-Unsupervised-Multitask-Learners-Radford-

Wu/9405cc0d6169988371b2755e573cc28650d14dfe), outperforms its counterparts, PLIP (https://www.biorxiv.org/content/10.1101/2023.03.29.534834v1), BiomedCLIP (https://arxiv.org/abs/2303.00915), and CLIP, in most datasets.

Results using cross-modal retrieval: In our study, we evaluate cross-modal retrieval efficacy by examining both zero-shot text-to-image and image-to-text retrieval capabilities. Our experiments are conducted on two datasets: our holdout dataset from Quilt-1M and the ARCH dataset. In terms of cross-modal capabilities our QuiltNet performs better in most of the cases compared to it's counterparts.

Links to our model demonstrations can be found here: QuiltNet-B-32 (https://huggingface.co/wisdomik/QuiltNet-B-32)

QuiltNet-B-16 (https://huggingface.co/wisdomik/QuiltNet-B-16)

QuiltNet-B-16-PMB (https://huggingface.co/wisdomik/QuiltNet-B-16-PMB)

Authors

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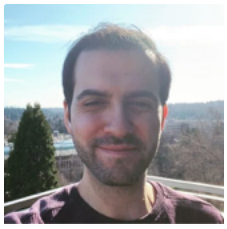

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Citation

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Contact

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